TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	TEER sensing functionality in organ on chip applications.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-electronics / Bio-Information	Medical Bio	Pharmaceutical Bio-based Technologies and Systems
Concept of Proposed Technology	The core concept is the development and integration of **biocompatible electrode** materials within **microfluidic organ-on-chip devices** to enable continuous or periodic TEER measurements without compromising optical access, tissue function, or experimental workflows. These electrodes will allow real-time or near-real-time electrical monitoring of tissue barrier properties within a **dynamic flow environment** while preserving compatibility with standard imaging and biochemical assays.			
Rationale for Joint R&D Project	The proposed objectives cannot be achieved through isolated development efforts. A joint R&D approach with organ-on-chip companies is essential for the following reasons: **Co-Development with Real-World Platforms** Organ-on-chip developers possess deep, application-specific knowledge of: Tissue architecture and biological constraints, Platform-specific fabrication processes, Performance requirements driven by end users (e.g., pharma). Collaborative development ensures that integrated electrode solutions are: • Directly compatible with existing and emerging OoC architectures • Validated under realistic biological and operational conditions • Designed for practical adoption rather than standalone demonstration			

	Bridging Disciplinary Boundaries The integration of electrodes for TEER measurement sits at the intersection of: • Advanced electrode materials & design and production processes • Electrical sensing and signal stability • Cell–material interactions • Microfluidic system design. No single entity can effectively optimize all dimensions independently. Collaboration enables iterative design, testing, and refinement across disciplines, reducing technical risk and accelerating innovation. **Accelerating Translation and Impact** Collaborative R&D allows outcomes to be: • Tested across multiple organ models and platforms • Assessed for scalability and manufacturability • Aligned with downstream regulatory and end-user expectations This increases the likelihood that project outputs will transition from research prototypes to widely usable technologies.
Key Words	1. Organ-on-chip platform 2. Real Time TEER measurement 3. Biocompatible electrodes integration 4. Flat panel display technology 5. Readout systems

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* 1. Biocompatibility of conductive electrode materials 2. Process feasibility of electrode integration in chips that allows optical access of tissue 3. TEER measurement in dynamic flow conditions 4. Development of scalable solution for TEER integration in Organ on chip platforms			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* 1. Influence of biocompatible electrodes on biological model ±10% of reference (without electrode) 2. Achieve target TEER measurement in static condition (depending upon biological model) 3. Achieve target TEER measurement in flow condition (depending upon biological model)			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The project will follow a stage-gated, risk-mitigated R&D approach, progressing from materials feasibility to lab scale validation and finally to a pilot prototype concept. Each phase concludes with defined technical milestones and Go/No-Go decision points to ensure scientific feasibility, biological compatibility, and translational relevance before advancing to subsequent stages. The approach emphasizes: • Early validation of biocompatibility, processability of selected electrode materials and signal stability • Strategies to combine TEER electrodes with optical access of tissue will be developed • Iterative co-design with the organ-on-chip partner, assuring optical access of tissue • Progressive testing under static and physiologically relevant flow conditions			

Key Tasks, Methodologies, and Milestones

WP1: Feasibility and Biocompatibility Assessment of Electrode Materials

Objective: Evaluate candidate electrode materials for suitability in organ-on-chip that allow optical access of barrier model.

Key Activities:

- Selection and fabrication of candidate scalable electrode materials
- Surface characterization (electrical conductivity, optical transparency, stability)
- In vitro biocompatibility testing with intestinal model
- Assessment of viability and barrier function compared to reference electrode materials
- Develop first system design concept that combines TEER electrodes with optical access of tissue

Methodologies:

- Standard cell viability tests
- Optical microscopy and fluorescence imaging
- Electrical stability testing in culture media

Milestone 1:

Demonstration of biocompatible electrodes with stable electrical properties

Go/No-Go Decision Point:

Proceed only if electrodes meet predefined criteria for:

- Cell viability and barrier function
- Optical compatibility
- Electrical stability over relevant timeframes

Indicative Timeline: Months 0–6

WP2: Conceptual Design of Electrode Integration into Organ-on-Chip Platforms

Objective:

Define how TEER electrodes can be structurally and functionally integrated into the partner's organ-on-chip design.

Key Activities:

- Co-design workshops with organ-on-chip partner
- Evaluation of electrode placement, geometry, and interfaces
- Assessment of integration constraints related to microfluidics, flow, and tissue imaging

Methodologies:

- CAD-based conceptual design
- Design-for-manufacturing considerations
- Risk analysis of integration strategies

Milestone 2:

Defined and agreed-upon electrode integration concept compatible with partner's OoC platform

Indicative Timeline: Months 6–8

WP3: Development of a Lab-Scale Integrated Prototype

Objective: Fabricate and validate a lab-based prototype incorporating TEER electrodes and electrical readout capability.

Key Activities:

- Prototype fabrication integrating electrodes into a test OoC device
- Electrical interfacing and signal acquisition setup
- Initial validation of signal detectability and reproducibility

Methodologies:

- Microfabrication and assembly techniques
- Electrical impedance measurement
- Bench-top functional testing

Milestone 3:

Functional lab prototype capable of electrode-based electrical signal readout
Indicative Timeline: Months 8–15

WP4: Development and Optimization of the TEER Measurement System

Objective: Establish reliable TEER measurement protocols using the integrated prototype.
Key Activities:

- Optimization of measurement parameters (frequency, voltage, signal processing)
- TEER measurements under static culture conditions
- TEER measurements under dynamic flow on both apical and basolateral sides of an intestinal barrier model
- Correlation of TEER data with biological readouts

Methodologies:

- Impedance spectroscopy
- Controlled microfluidic perfusion
- Comparative analysis between static and dynamic conditions

Milestone 4:
Demonstration of stable, reproducible TEER measurements under static and dynamic flow conditions
Go/No-Go Decision Point:
Advance only if TEER measurements:

- Are stable over time
- Respond appropriately to barrier formation and disruption
- Are compatible with dynamic flow conditions

Indicative Timeline: Months 15-21

WP5: Definition of Pilot Prototype Concept

Objective: Translate laboratory findings into a scalable, application-relevant pilot prototype concept.
Key Activities:

- Analysis of lab prototype performance and limitations
- Definition of functional and technical requirements for a pilot prototype
- Identification of scalability, manufacturability, and usability considerations

Methodologies:

- Design reviews and performance benchmarking
- Stakeholder input from organ-on-chip partner
- Risk and feasibility assessment

Milestone 5:
Defined pilot prototype concept informed by validated lab-scale results
Indicative Timeline: Months 21-24

WP6: Pilot production and Performance Evaluation

Objective: Produce pilot-scale prototypes and evaluate their technical and biological performance under realistic operating conditions.
Key Activities:

- Production of a limited series of pilot prototypes
- Verification of manufacturing reproducibility and assembly robustness
- Performance evaluation of integrated electrodes and TEER measurement system
- Assessment of usability and integration within standard experimental workflows

Methodologies:

- Pilot-scale manufacturing and quality checks
- Electrical and optical performance testing
- TEER monitoring
- Comparative evaluation against lab-scale results

Milestone 6:
Validated pilot prototype demonstrating reproducible performance and operational robustness

	Go/No-Go Decision Point: Decision on readiness for broader deployment, further optimization, or scale-up. Indicative Timeline: Months 25–36

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*
	TNO will serve as the lead scientific and technical contributor for the development and validation of biocompatible electrode technologies and TEER measurement methodologies.
	
	In general, TNO will be responsible for:
	• Research and development of biocompatible electrode materials
	• Design and optimization of electrical sensing and TEER measurement methodologies
	• Development of signal acquisition, processing, and analysis protocols
	• Establishment of performance metrics, benchmarks, and validation criteria
	
	Specific TNO responsibilities:
	• Electrode Feasibility and Biocompatibility
	  ○ Selection, fabrication, and characterization of candidate biocompatible electrode materials
	  ○ In vitro biocompatibility and stability testing
	  ○ Definition of Go/No-Go criteria for electrode suitability
	• Electrical System Development
	  ○ Design and optimization of TEER measurement strategies (frequency selection, signal stability)
	  ○ Development of laboratory measurement setups and data acquisition workflows
	• Lab-Scale Prototype Development
	  ○ Fabrication and testing of lab-based integrated prototypes in collaboration with the OoC partner
	  ○ Electrical validation and performance assessment of integrated electrodes
	• Experimental Design and Data Analysis
	  ○ Design of static and dynamic TEER experiments
	  ○ Data interpretation and correlation with biological readouts
	• Pilot Evaluation Support
	  ○ Definition of quality protocols for pilot prototype production
	
	TNO will provide:
	• Specialized laboratories, cleanrooms and equipment for scalable flat panel display compatible manufacturing, electrical testing, and cell culturing
	• Expertise in:
	  ○ Materials science & Bioelectronics
	  ○ Scalable manufacturing
	  ○ Microfluidics and impedance spectroscopy
	  ○ Data analysis tools and documentation of methodologies
	  ○ Project coordination and technical reporting support
	
	Collaboration Responsibilities
	• Active participation in joint design reviews and milestone evaluations
	• Transparent sharing of data, results, and lessons learned within the project framework

	• Contribution to jointly developed documentation, guidelines, and reports

Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*
	The organ-on-chip company will act as the application and platform development partner, providing the biological, microfluidic, and manufacturing expertise necessary to translate the electrode and TEER technologies into a functional and relevant organ-on-chip system.
	
	In general, the OoC company will be responsible for:
	• Providing organ-on-chip platform designs, specifications, and fabrication expertise
	• Ensuring biological relevance and functional performance of the tissue models
	• Supporting prototype integration, testing, and iteration
	• Leading pilot prototype production and manufacturability assessment
	
	Specific OoC company responsibilities:
	• Platform Integration
	  ○ Provision of chip designs, materials, and fabrication processes
	  ○ Co-development of electrode integration concepts compatible with existing platforms
	  ○ Assembly of integrated lab-scale and pilot prototypes
	• Biological Model Development
	  ○ Establishment and maintenance of intestinal barrier tissue models
	  ○ Execution of cell culture, differentiation, and barrier formation protocols
	  ○ Support for long-term culture and dynamic flow experiments
	• Testing and Validation
	  ○ Biological validation of TEER measurements under static and dynamic conditions
	  ○ Comparative assessment of barrier integrity using complementary assays
	  ○ Participation in Go/No-Go milestone evaluations
	• Pilot Prototype Production
	  ○ Production of a limited batch of pilot prototypes
	  ○ Assessment of reproducibility, robustness, and usability
	  ○ Feedback on scalability and manufacturing constraints
	
	The OoC company will provide:
	• Access to organ-on-chip fabrication facilities and tooling
	• Biological laboratories and cell culture resources
	• Dedicated engineering and biological personnel
	• Prototype materials, consumables, and test platforms
	
	Collaboration Responsibilities
	• Active participation in joint design reviews and milestone evaluations
	• Timely sharing of platform data, performance feedback, and design constraints
	• Contribution to joint dissemination activities, where appropriate
	• Jointly define project milestones, success criteria, and Go/No-Go decision points

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*
	Core Technological Achievements
	• Validated electrode technology that is biocompatible and provides optical access, and is also electrically stable under long-term cell culture and dynamic flow conditions
	• Integrated TEER measurement capability embedded directly within organ-on-chip platforms, enabling non-invasive, real-time or periodic monitoring of tissue barrier integrity
	• Optimized TEER measurement methodologies, including frequency selection, signal processing, and data interpretation, applicable to both static and dynamic culture conditions
	• Lab-scale and pilot-scale prototypes demonstrating reliable performance, reproducibility, and usability
	The outcomes of this joint R&D project are expected to directly support the commercial growth, product differentiation, and market competitiveness of the participating Korean SME (OoC company).
	• Product Differentiation and Value Creation
	  ○ Integration of embedded TEER measurement will enable the SME to offer organ-on-chip platforms with built-in, real-time tissue quality monitoring, a feature not widely available in current commercial solutions.
	  ○ TEER-enabled platforms will provide clear added value to end users by enabling continuous assessment of barrier integrity, tissue maturation, and long-term viability.
	• Enhanced monitoring capabilities will increase the attractiveness of the SME's platforms for:
	  ○ Pharmaceutical and biotechnology companies
	  ○ Contract research organizations (CROs)
	  ○ Academic and translational research institutes
	  ○ TEER-enabled systems support advanced applications such as chronic toxicity testing and permeability studies, allowing the SME to enter higher-value and more demanding market segments.
	• Reduced Development and Commercial Risk
	  ○ Validation of the technology through a joint, stage-gated R&D process reduces technical uncertainty prior to commercialization.
	  ○ Pilot-scale production and performance evaluation provide early insight into manufacturability, reproducibility, and usability, lowering barriers to market entry.
	Broader Economic and Societal Impact
	• Economic and Industrial Impact
	  ○ Strengthening the technological capabilities and global competitiveness of Korean SMEs in the organ-on-chip and bioengineering sectors
	  ○ Supporting the development of high-value, knowledge-intensive products with export potential
	• Impact on the Life Sciences and Biomedical Research Ecosystem
	  ○ Improved reliability and reproducibility of organ-on-chip experiments through continuous tissue quality monitoring
	  ○ Facilitation of longer-term and more physiologically relevant studies, accelerating adoption of OoC technologies across research and industry
	  ○ Contribution to pre-competitive standards and best practices for integrated sensing in OoC systems
	• Societal and Public Health Impact
	  ○ Support for more predictive, human-relevant preclinical testing, contributing to:
	  ▪ Improved drug safety and efficacy assessment
	  ▪ Reduced reliance on animal testing
	  ○ Long-term potential to lower drug development costs and timelines, benefiting patients and healthcare systems

6. References

(1) Griep, L.M., Wolbers, F., de Wagenaar, B. *et al.* BBB ON CHIP: microfluidic platform to mechanically and biochemically modulate blood-brain barrier function. *Biomed Microdevices* 15, 145–150 (2013). https://doi.org/10.1007/s10544-012-9699-7

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization